

MICRO-DEGREE

Künstliche Intelligenz und Gesellschaft

***Artificial Intelligence
and Society***



Micro-Degree Artificial Intelligence and Society

– Technical Aspects of AI –

Lecture 1.2: Supervised Learning

Jana Lasser

Research group

Complex Social & Computational Systems (CS)²

IDEa_Lab

Das Interdisziplinäre Digitale
Labor der Universität Graz



Supervised Learning: Classification

- A number of *observations* with *attributes* and their *values* ($x_1, \dots, x_d$).
- For each observation, a class (label) $y \in \{y_1, \dots, y_k\}$.

Goal: Find the correct class for a new observation *without a class*.

→ **Classification**

Example

y_1 : Horse



y_2 : Pony



Sources: pxhere.com, [Wikimedia commons](https://commons.wikimedia.org), CC BY 2.0

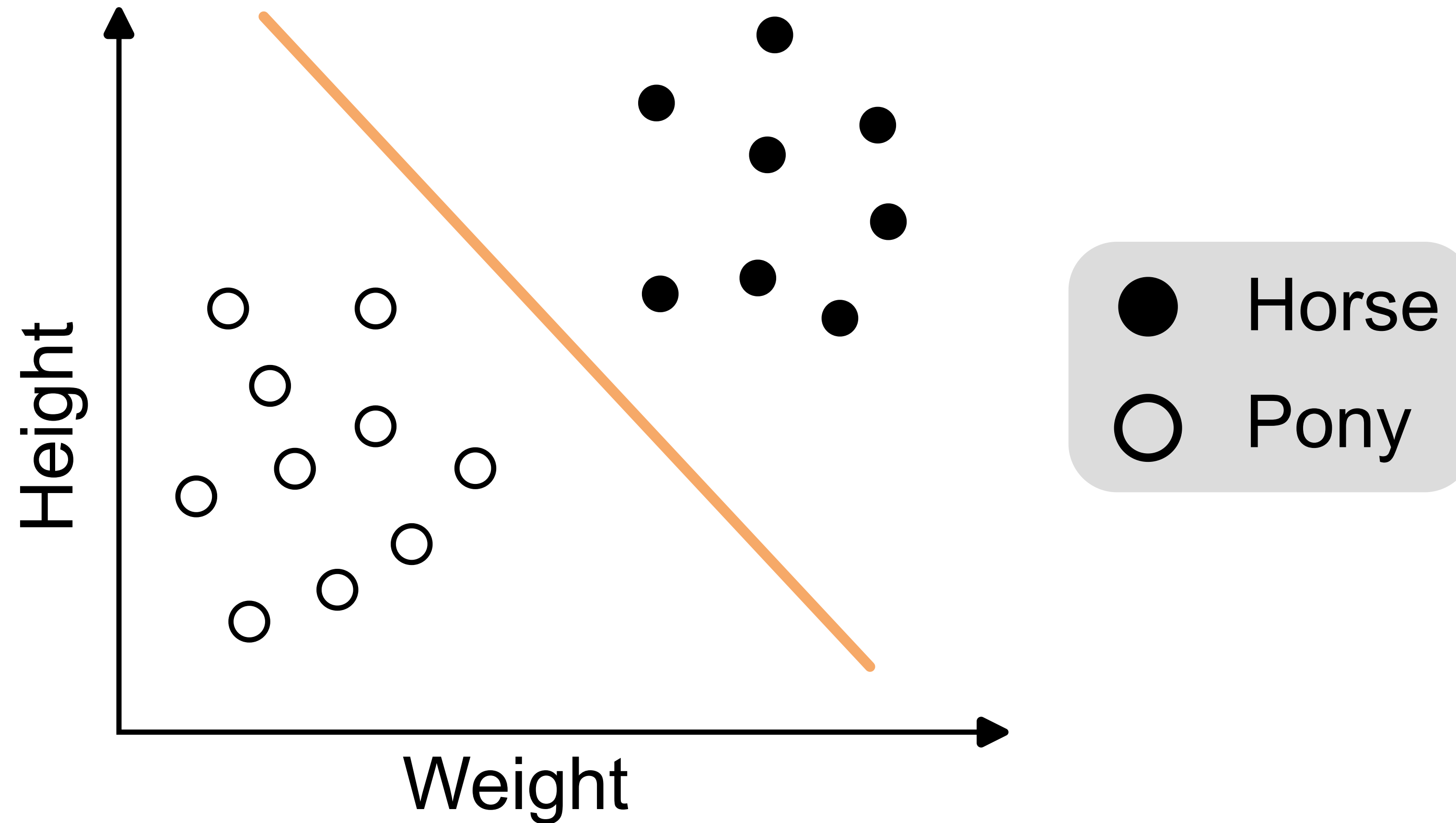
- x_1 : Weight
- x_2 : Height

Observation: Animal weighing 520 kg and 148 cm tall.

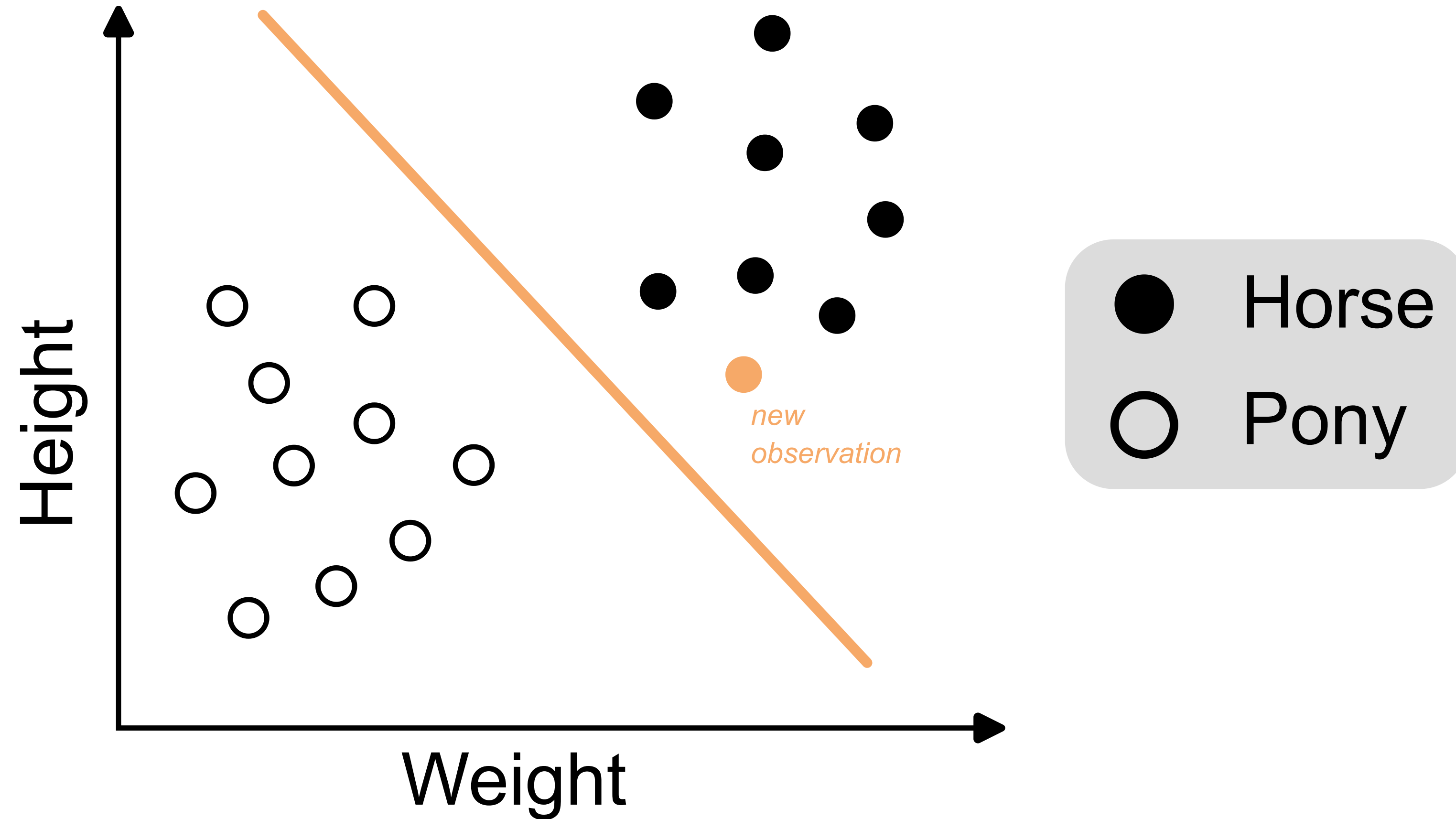
→ Horse

Supervised Learning: Classification

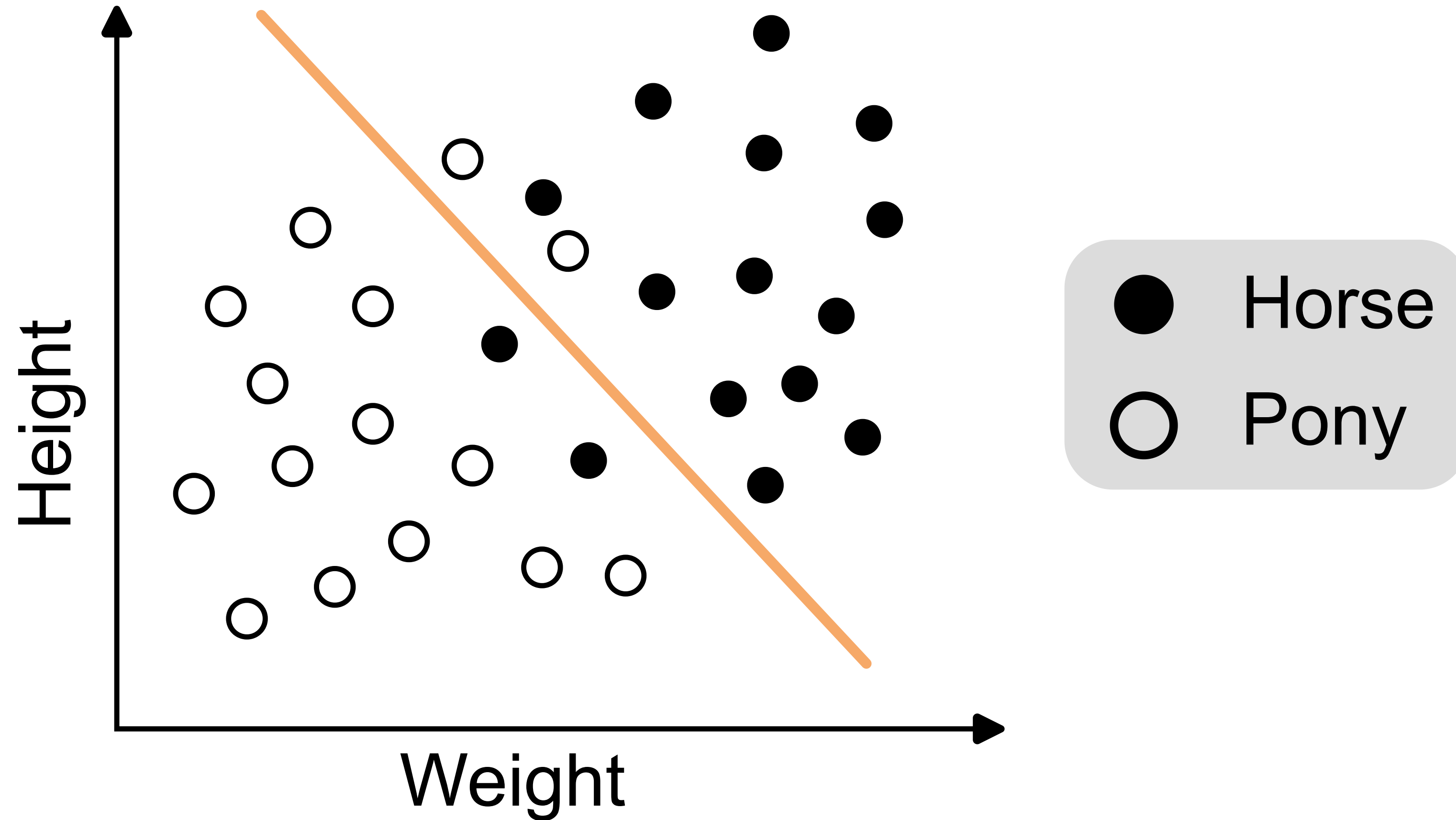
Find a line that optimally separates horses and ponies.



Supervised Learning: Classification

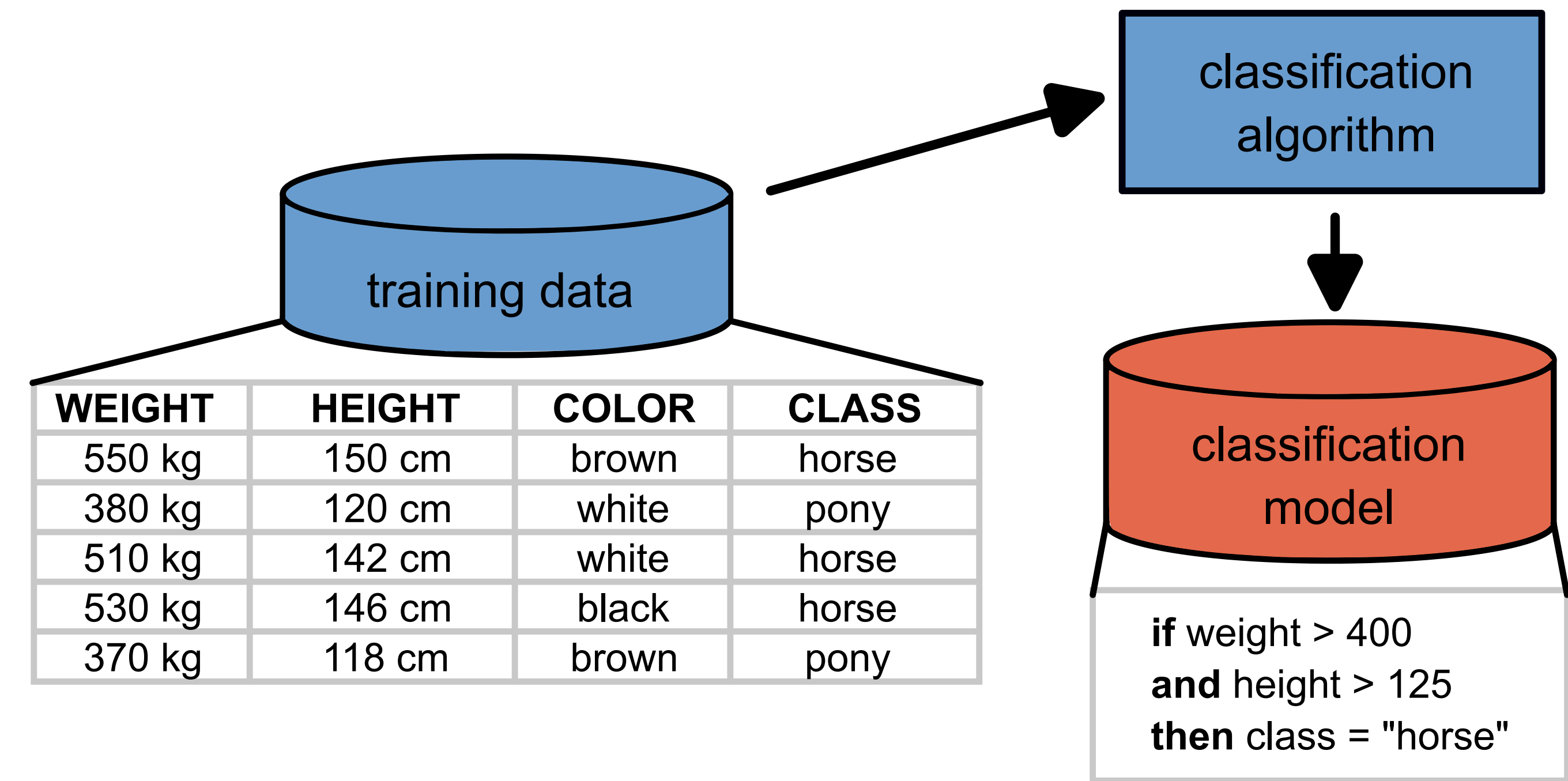


Supervised Learning: Classification

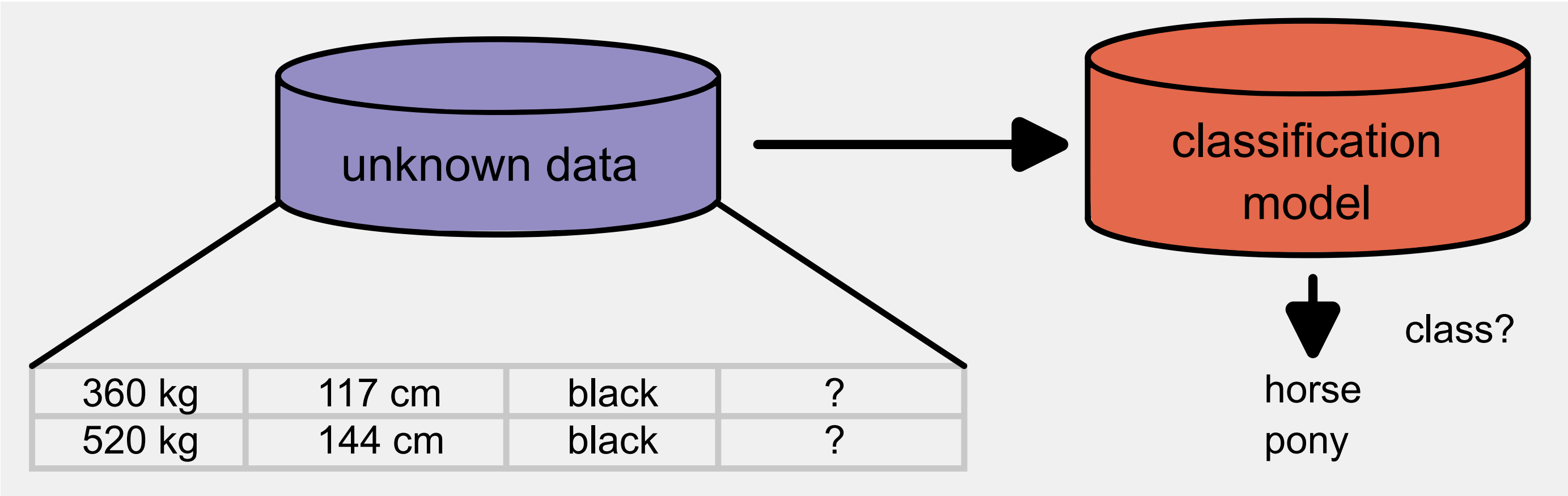


Classification "Pipeline"

Training Phase



Application Phase



Summary

- In **supervised learning**, we try to learn known patterns and recognize them automatically.
- Supervised learning consists of a **training phase** and an **application phase**.
- With supervised learning, we can predict class membership (**classification**).
- Supervised learning also allows predicting continuous values (**regression**) → more in the next lecture.
- Nomenclature:
 - **Attributes** (like "weight" and "height" in the animal example) are also often called **features**.
 - **Classes** (like "horse" and "pony" in the animal example) are also often called **labels** or **target values**.
 - **Observations** (like triplets of weight, height and class in the animal example) are also often called **data points**.

Exercise (5 min)

- Think of a **use case for supervised learning** from your daily life, e.g., from studies, work, or private life.
- For the given use case, consider:
 - What is a single **observation** (or "data point")?
 - What are the **attributes** (or "features") of the observation?
 - What are the **classes** (or "labels", "target values") you want to predict?